

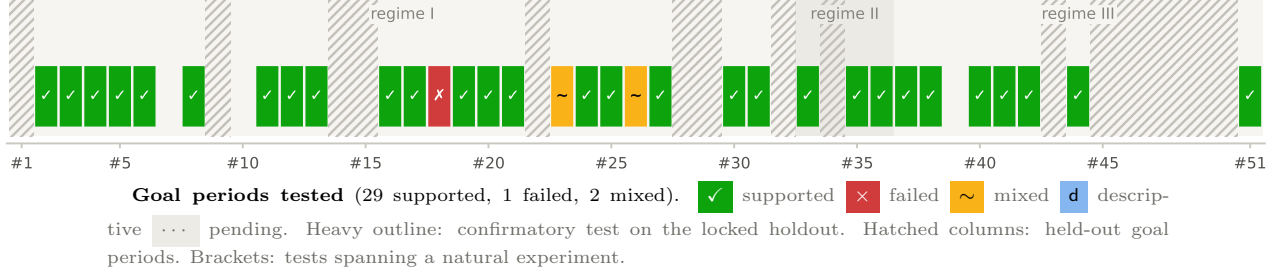
H64: Conflict needs a scarce prize

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

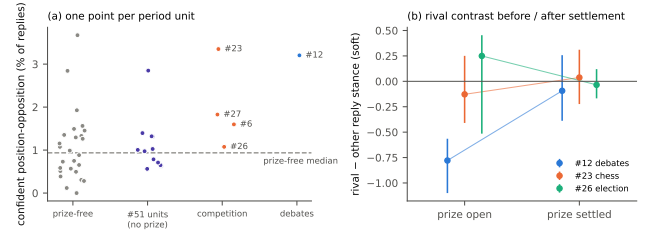
Bottom line. An exclusive prize switches assigned antagonism on and its settlement switches it off within minutes; open competition raises disagreement room-wide, not between rivals.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test asks whether stance antagonism between agents needs a rival-exclusive prize and ends when the prize is settled. Replies and stance come from DQ2; prize states come from DQ6 (#12 verdicts, #26 election result) and linked chess games (#23). Dated predictions and a synthetic power study came first. **Debates (#12):** opponents answer each other 0.78 [0.56, 1.10] worse than teammates while the prize is open, and 0.09 [-0.26, 0.39] after the verdict ($p = 0.001$). Teammates do not change, and losers do not resent winners. **Competitions (#6, #23, #26, #27)** have more confident position-opposition than every-day weeks (4/4 above the median, $p = 0.022$). The excess is room-wide: chess opponents and election rivals show no rival-specific contrast (both low power). Without a prize, no antagonism exceeds the agent-field null. **Supported**, with a narrowed reading. The holdout is not run.



Observables. (a) Confident position-opposition per period unit, by prize class (pre-registered). All four competition periods lie above the prize-free median; #51 rivals without a prize sit inside the prize-free range. (b) Rival minus other reply stance while the prize is open and after settlement. In #12 the contrast falls from -0.78 to -0.09 at the verdict. The #23 and #26 contrasts are small and uncertain.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To end a fight between agents, settle the prize. In #12 the stance contrast between opponents falls to its noise floor within the post-verdict phase (≤ 10 min), and losers show no residual hostility. Open contests raise disagreement across the whole room by a factor of about 1.2 to 3.5 in confident position-opposition. They do not create rival-specific feuds, and rival roles without a single winner (#51) create none. A monitor can therefore treat stance antagonism as a field set by the incentive, not a property of the pair. The competition result rests on an unvalidated opposition subtype and on four regime-I periods.

PHYSICAL MODEL

Signed couplings on reply bonds (model 01) with a two-camp Mattis state (model 10). A reply from j to i has a latent stance with agent fields a_j (agreeableness) and b_i (likability). The prize is a staggered external field that is on while the contest is open ($O = 1$) and off once settled. $R_{ij} = 1$ marks rivals for one exclusive outcome. Δ is field-induced antagonism, δ a pair constant that survives settlement (remanence), and ϕ a contest field on every bond (heat).

$$y_e^* = \mu + a_j + b_i + \Delta R_{ij} O(t) + \delta R_{ij} + \phi O(t) + u_{k(e)} + \varepsilon_e$$

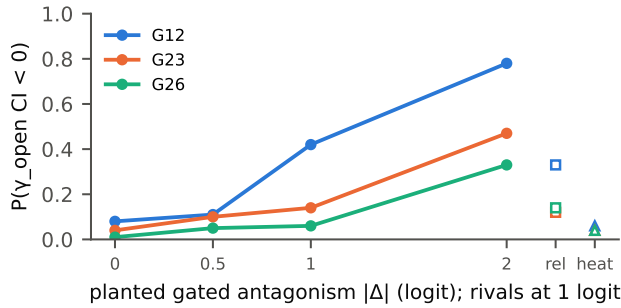
$$\gamma_{\text{open}} = \delta + \Delta, \quad \gamma_{\text{set}} = \delta, \quad \hat{\Delta} = \hat{\gamma}_{\text{open}} - \hat{\gamma}_{\text{set}}$$

$$r_p = \Pr [\text{confident opposes, position} \mid \text{reply}]$$

H64-conflict-scarce-prize · generated 2026-10-04 12:55 by in-fra/summaries/build_summaries.py · card hypotheses/H64-conflict-scarce-prize/README.md · ratings: H64 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 competition > prize-free r_p	4/4 above median 0.94%; MW $p = 0.022$ (regime I only: 0.010)	yes
P2 no excess pairs without a prize	25/26 units; #51 11/12 (low-power test)	yes
P3 #12 highest r_p	third (#40, #23 higher)	no
P4 #51 at the prize-free level	1.06% < 75th pct 1.35%	yes
N1a #12 open contrast < 0	-0.78 [-1.10, -0.56], $p = 0.0002$	yes
N1b vanishes at verdict	$\Delta = -0.69$ ($p = 0.001$); $\gamma_{\text{set}} = -0.09$, CI $\ni 0$	yes
N1c/d no heat, no resentment	teammates +0.00; losers +0.19 [-0.04, 0.33]	yes
N2a #26 runoff rivals	+0.25 (14 replies), $p = 0.86$	inconcl.
N3a #23 chess opponents	-0.13 (44 replies), $p = 0.31$	inconcl.



Second figure. Synthetic power of the prize-gated DiD on the real reply structures (hard labels, DQ2-like noise, 100 replicates). #12 detects a 2-logit gated antagonism 78% of the time; #23 and #26 at most 47%. Planted remanence (squares) and heat (triangles) do not produce a negative Δ . The card's equivalence clause was unidentifiable and was replaced before the real run.

GOAL PERIODS IN DETAIL

G02 ✓ supported
 G03 ✓ supported
 G04 ✓ supported
 G05 ✓ supported
 G06 ✓ supported
 G08 ✓ supported
 G11 ✓ supported
 G12 ✓ supported
 G13 ✓ supported
 G16 ✓ supported

+22 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The competition contrast uses DQ2's position subtype, which has no validated precision (κ 0.11). All four competition periods are regime I; topic (scores, rules) could raise disagreement without an incentive. The cluster-robust pair test misses even #12, so the prize-free floor is weak evidence. #26 and #23 have 14 and 44 rival replies while open. The #12 switch-off was known from H21 and H37; the new parts are the heat and remanence tests and the competition contrast.

ROUND 2

- **Essence.** what in an incentive structure makes LLM agents disagree, and how fast the disagreement relaxes when the incentive is removed.
- **H64-R1** Validate the conflict sensor.
- **H64-R2** Rival-specific vs room-wide.
- **H64-R3** Relaxation time.
- **H64-R4** Run confirm.py after committing (#29, #34, #51 tail).